

Course Name:	Computer Networks	Course Code:	CS 3001
Program:	BS (Computer Science)	Semester:	Spring 2026
Duration:	15 minutes	Total Marks:	15
Paper Date:		Section	6B1
Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

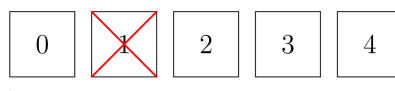
Q1. Carefully read the Question before attempting:

[5 marks]

Consider a pipelined protocol over a link with a very high Bandwidth-Delay Product. The sender sends packets 0, 1, 2, 3, 4. Packet 1 is lost. All other packets and ACKs arrive correctly. Timeout occurs after Packet 4 is sent. Window Size $N = 5$. Sequence Numbers: 0, 1, 2, ...

- If using **Go-Back-N**, mathematically calculate the total number of packet transmissions (initial + retransmissions) required to successfully deliver packets 0 through 4. If using **Selective Repeat**, calculate the same.
- Will **Selective Repeat** operate correctly if $N = 5$ and $k = 3$? Consider the SR window size constraint, create a scenario where it fails and explain it (why)?

Lost

Window $N = 5$

Q2. Carefully read the Question before attempting:

[10 marks]

Host A sends 4 segments to Host B over a TCP connection.

- Segment 1: Seq = 90, Length = 20 bytes.
- Segment 2: Seq = 110, Length = 40 bytes.
- Segment 3: Seq = 150, Length = 10 bytes.
- Segment 4: Seq = 160, Length = 30 bytes.

Assume the initial Sequence Number is 90. Host B receives them in this order: Segment 1, Segment 4, Segment 2, Segment 3.

- List the ACK number Host B sends immediately after receiving EACH segment in that specific order.
- Draw the receive buffer state after Segment 4 arrives but before Segment 2 arrives.

